

L -functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

David A. McGady

Abstract

We give a single finite, regulator-free functional realizing in one object the period polynomials (at integer s) and the completed L -functions (at general s) of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$, with poles anywhere: at the cusp and in the interior. It is a two-segment contour integral with an explicit Hurwitz-zeta T -kernel, built from any finite base point $\tau_0 \in \mathbb{H}$ and manifestly finite. For weakly holomorphic forms it is independent of τ_0 and of the path, reproducing at $\tau_0 = i$ the special L -values of Bringmann–Fricke–Kent [1]; its integer- s values give the period polynomial, Brown’s quasi-periods [2] in closed form. For forms with interior poles the value depends only on the homotopy class of the contour — fixed up to the residues it crosses — and a natural choice of class, the one prescription a pole on the line $\mathrm{im} \tau = 1$ requires, renders it canonical. The resulting closed form generalizes [3]: a regularized cusp sum plus one polylog integral per interior pole. The construction introduces no new objects, unifying these periods and L -values through a single all- s kernel, and every formula is checked numerically [4].

1 A finite L -integral

Definition 1.1 (S - and T -segments). *Let $\tau_0 \in \mathbb{H}$ and $f \in F_k$, and let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ generate $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. The S - and T -segments at τ_0 are smooth paths in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$,*

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0, \quad (1)$$

running from $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 .

The pole-avoidance is inert for $f \in M_k^!$: with no interior poles, $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty) = \mathbb{H}$ and the segments are unobstructed. Interior poles — and the homotopy classes of contours they create — become relevant only for meromorphic f , in the final section.

Definition 1.2 (Hurwitz T -kernel). *The Hurwitz T -kernel is, with $\zeta(s, x) = \sum_{m \geq 0} (m + x)^{-s}$ the Hurwitz zeta function (meromorphic in s , holomorphic in $x \in \mathbb{H}$),*

$$\tilde{k}_T(\tau, s) := \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(s - (k-1), \tau + 1). \quad (2)$$

Definition 1.3 (L -integral). *Let $f \in F_k$, $\tau_0 \in \mathbb{H}$, and $s \in \mathbb{C}$, and let $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ be the S - and T -segments at τ_0 , as in Definition 1.1. The L -integral of f is*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right]. \quad (3)$$

Its τ_0 - and path-dependence is dealt with below, in the main body.

Definition 1.4 (*L-function*). *The L-function of $f \in F_k$ is*

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (4)$$

Likewise; once that dependence is settled below, we drop the arguments and write simply $L^*(f, s)$ and $L(f, s)$.

Lemma 1.5 (Finiteness). *For $f \in F_k$, both integrals in Definition 1.3 are finite for every $s \in \mathbb{C}$, every $\tau_0 \in \mathbb{H}$, and every choice of paths $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$.*

Proof. Both segments are compact and lie in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so f is continuous on them; τ^{s-1} is continuous on $\mathbb{H}$, and $\tilde{k}_T(\tau, s)$ is continuous there since $\zeta(\nu, \tau + 1)$ is holomorphic in $\tau \in \mathbb{H}$ (Definition 1.2). A continuous integrand on a compact path is integrable, for every $s \in \mathbb{C}$. $\square$

Proposition 1.6 (Basepoint and path independence). *For $f \in M_k^!$ the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of $\tau_0 \in \mathbb{H}$ and of the paths: with no interior poles, $\mathbb{H}$ is simply connected, so all admissible contours are homotopic. We write $L^*(f, s)$ for the common value and $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$.*

Proof. With no interior poles the integrands $f(\tau)\tau^{s-1}$ and $f(\tau)\tilde{k}_T(\tau, s)$ are holomorphic on $\mathbb{H}$, which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of τ_0 we differentiate. From $\zeta(\nu, x) = \sum_{m \geq 0} (m+x)^{-\nu}$ one has $\zeta(\nu, x+1) - \zeta(\nu, x) = -x^{-\nu}$; applied to both terms of the kernel at $x = \tau + 1$,

$$\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s}. \quad (5)$$

The T -segment runs $\tau_0 - 1 \rightarrow \tau_0$, so *both* endpoints move with τ_0 ; by Leibniz's rule and the T -periodicity $f(\tau_0 - 1) = f(\tau_0)$,

$$\partial_{\tau_0} \int_{\gamma^T} f \tilde{k}_T = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (6)$$

The S -segment runs $-1/\tau_0 \rightarrow \tau_0$; its lower endpoint moves with $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$, so by Leibniz, the modularity $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, and the phase $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{1-s}$,

$$\partial_{\tau_0} \int_{\gamma^S} f \tau^{s-1} = f(\tau_0) \tau_0^{s-1} - f(-1/\tau_0) (-1/\tau_0)^{s-1} \tau_0^{-2} = f(\tau_0) [\tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (7)$$

The two brackets are negatives of each other, so $\partial_{\tau_0} L^* = i^{-s} (\partial_{\tau_0} \int_{\gamma^S} + \partial_{\tau_0} \int_{\gamma^T}) = 0$. $\square$

Remark 1.7 (Classical Mellin recovery). *For a holomorphic cusp form $f \in S_k$, the cusp limit $\tau_0 \rightarrow i\infty$ collapses the S -segment onto $[0, i\infty]$ while the T -segment vanishes exponentially, so*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt, \quad (8)$$

the classical completed L-integral.

2 The L -function

Proposition 2.1 (Cusp unfolding). *For $n > 0$ and $\nu \in \mathbb{C}$,*

$$\int_i^{i+1} e^{2\pi i n u} \zeta(\nu, u) du = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (9)$$

Proof. Write $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$ and shift $u \mapsto u+m$ in the m -th term; as $e^{2\pi i n(u+m)} = e^{2\pi i n u}$ for integer m , the unit segments stack into one ray,

$$\int_i^{i+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{i+m}^{i+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_i^{i+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (10)$$

For $n > 0$ the integrand is holomorphic in the first quadrant and decays as $\text{im } v \rightarrow \infty$ (since $|e^{2\pi i n v}| = e^{-2\pi n \text{im } v}$), so the horizontal ray rotates to the vertical $[i, i\infty)$. With $v = it$ ($t \geq 1$; then $dv = i dt$, $e^{2\pi i n v} = e^{-2\pi n t}$, $v^{-\nu} = i^{-\nu} t^{-\nu}$) and $w = 2\pi n t$,

$$\int_i^{i+\infty} e^{2\pi i n v} v^{-\nu} dv = i^{1-\nu} \int_1^\infty e^{-2\pi n t} t^{-\nu} dt = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (11)$$

□

Theorem 2.2 (L -function for weakly holomorphic forms). *For $f \in M_k^!$, at $\tau_0 = i$,*

$$L^*(f, s) = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right] = i^k L^*(f, k-s). \quad (12)$$

Proof. At $\tau_0 = i$ the S -segment degenerates ($-1/i = i$), so $L^*(f, s) = i^{-s} \int_{i-1}^i f \tilde{k}_T d\tau$. Expand $f = \sum_n c_f(n) e^{2\pi i n \tau}$ and substitute $u = \tau + 1$ ($e^{2\pi i n(u-1)} = e^{2\pi i n u}$): each $n > 0$ mode is a cusp unfolding (Proposition 2.1), and the closed form persists to the finitely many $n < 0$ modes (the cusp principal part) by analytic continuation. The first kernel term ($\nu = 1-s$, so $1-\nu = s$) gives $i^{-s} i^s (2\pi n)^{-s} \Gamma(s, 2\pi n) = \Gamma(s, 2\pi n)/(2\pi n)^s$; the second ($\nu = s-k+1$, so $1-\nu = k-s$) gives

$$-i^{-s} e^{i\pi(s-1)} i^{k-s} (2\pi n)^{s-k} \Gamma(k-s, 2\pi n) = i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}, \quad (13)$$

the s cancelling in the phase $-i^{-s} e^{i\pi(s-1)} i^{k-s} = i^k$. So each $n \neq 0$ mode contributes $\Gamma(s, 2\pi n)/(2\pi n)^s + i^k \Gamma(k-s, 2\pi n)/(2\pi n)^{k-s}$, while the $n = 0$ mode contributes $c_f(0) i^{-s} \int_{i-1}^i \tilde{k}_T d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right)$. Swapping the two Γ -terms and using $i^{2k} = (-1)^k = 1$ (the weight k is even) shows that bracket equals i^k times its $s \mapsto k-s$ image; the polar term is i^k -symmetric likewise, giving $L^*(f, s) = i^k L^*(f, k-s)$. □

Definition 2.3 (Period cocycle). *Let $V_n := \mathbb{C}[X, Y]_n$ with the action $(P|_\gamma)(X, Y) := P(aX + bY, cX + dY)$, and $\omega_f := f(\tau) (X - \tau Y)^n d\tau$. For $f \in F_k$ and $\tau_0 \in \mathbb{H}$, the period cocycle is*

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f \in V_n, \quad \gamma \in \Gamma, \quad (14)$$

satisfying $C_{gh}^f = C_g^f|_h + C_h^f$.

Proof of the cocycle relation. Split the contour of C_{gh}^f at $h^{-1}\tau_0$: $\int_{h^{-1}g^{-1}\tau_0}^{\tau_0} \omega_f = \int_{h^{-1}g^{-1}\tau_0}^{h^{-1}\tau_0} \omega_f + \int_{h^{-1}\tau_0}^{\tau_0} \omega_f$. The second term is C_h^f ; by the Γ -equivariance of ω_f ($f|_k h = f$ gives $\int_{h^{-1}a}^{h^{-1}b} \omega_f = (\int_a^b \omega_f)|_h$) the first is $C_g^f|_h$, whence $C_{gh}^f = C_g^f|_h + C_h^f$. $\square$

Lemma 2.4 (Parabolic representatives exist iff $c_f(0) = 0$). *With $\delta_T P := P|_T - P$ on V_n , one has $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} c_f(0)$, so $C_T^f \in \text{im } \delta_T$ if and only if $c_f(0) = 0$ — the principal part of f never obstructs.*

Proof. The X^n -coefficient of ω_f is $f(\tau) d\tau$, so $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} f d\tau = (2\pi i)^{n+1} c_f(0)$, the $m \neq 0$ Fourier modes integrating to zero over a unit period. On the other side, $(P|_T)(X, Y) = P(X+Y, Y)$ has the same X^n -coefficient as P , so $[\delta_T P]_{X^n} = 0$ for every $P \in V_n$; since $\ker \delta_T = \mathbb{C} Y^n$ is one-dimensional, $\text{im } \delta_T$ is the entire hyperplane $\{Q : [Q]_{X^n} = 0\}$. Hence $C_T^f \in \text{im } \delta_T$ iff $[C_T^f]_{X^n} = 0$ iff $c_f(0) = 0$. $\square$

Definition 2.5 (Period polynomial). *For $f \in S_k^!$ (so $c_f(0) = 0$), the period polynomial is the S -value of the parabolic representative,*

$$r_f := C_S^f(\tau_0) - (P|_S - P) \in V_n, \quad (15)$$

with $P \in V_n$ chosen so that $\delta_T P = C_T^f$ (Lemma 2.4).

Proposition 2.6 (Period coefficients are special L -values). *For $f \in S_k^!$ and $\ell \in \{1, \dots, k-3\}$,*

$$[r_f]_{X^{n-\ell} Y^\ell} = (2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1). \quad (16)$$

Proof. The cocycle is the coboundary of the Eichler primitive $\tilde{F}(\tau) = \int^\tau f(z)(X - zY)^n dz$, namely $C_\gamma^f = \tilde{F} - \tilde{F}|_\gamma$, and $r_f = C_S^f - (P|_S - P)$ is the S -value of its cusp-regular (T -flat) branch, P solving $\delta_T P = C_T^f$ (Lemma 2.4). Read the $X^{n-\ell} Y^\ell$ -coefficient. From $(X - \tau Y)^n = \sum_j \binom{n}{j} (-\tau)^j X^{n-j} Y^j$,

$$[C_S^f]_{X^{n-\ell} Y^\ell} = (2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} \int_{\gamma_S} f \tau^\ell d\tau. \quad (17)$$

The kernel realizes the correction: $\tilde{k}_T(\tau, \ell+1)$ is the antidifference $\delta_T \tilde{k}_T(\tau, \ell+1) = -(\tau+1)^\ell + (-1)^\ell (\tau+1)^{n-\ell}$, whose two monomials are the ℓ - and $(n-\ell)$ -components exchanged by $Y|_S = X$; so the T -segment integral against $\tilde{k}_T$ is the coboundary P inverting δ_T , with $[P|_S - P]_{X^{n-\ell} Y^\ell} = -(2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} \int_{\gamma_T} f \tilde{k}_T(\tau, \ell+1)$. Therefore

$$[r_f]_{X^{n-\ell} Y^\ell} = (2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} \left[\int_{\gamma_S} f \tau^\ell + \int_{\gamma_T} f \tilde{k}_T(\tau, \ell+1) \right] = (2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1), \quad (18)$$

using $L^*(f, \ell+1) = i^{-(\ell+1)} [\int_{\gamma_S} f \tau^\ell + \int_{\gamma_T} f \tilde{k}_T(\tau, \ell+1)]$. $\square$

Remark 2.7 (Quasi-periods and their determinant). *For the weights with $\dim S_k = 1$ — $k \in \{12, 16, 18, 20, 22, 26\}$ — and a weak cusp form $f \in S_k^!$, the coefficients of r_f (Proposition 2.6) are Brown's quasi-period coordinates $\omega^\pm(f)$ [2]: ω^+ from ℓ even, ω^- from ℓ odd. Taking $f = \hat{\Delta}_k := q^{-1} + O(q^{\dim S_k+1}) \in S_k^!$, the canonical Duke–Jenkins weak cusp form [5] ($= q^{-1} + O(q^2)$ here, as $\dim S_k = 1$), they reproduce Brown's published $\eta^\pm(\hat{\Delta}_{12})$ at $k = 12$ and furnish the quasi-periods anew at the remaining five weights (Table 1). Their determinant satisfies a Legendre-type relation, $\omega^+ \eta^- - \omega^- \eta^+ \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$, equal to $-(k-2)! (2\pi i)^{k-1}$ at $k = 12$ in Brown's normalization (the weight- k analog of $\omega_1 \eta_2 - \omega_2 \eta_1 = 2\pi i$). Both are global consistency checks, claiming nothing new [4].*

k	$\omega^+(f_k)$	$\omega^-(f_k)/i$	$\eta^+(\widehat{\Delta}_k)$	$\eta^-(\widehat{\Delta}_k)/i$	$\det / (2\pi i)^{k-1}$
12	-6.89168e7	-5.58502e6	1.27202e11	1.02767e10	-3628800
16	-8.81481e11	-1.54063e10	4.14120e13	6.83541e11	-37777259520
18	-5.07833e13	-5.43305e12	-2.19200e14	-3.36339e13	13948526592000
20	-4.22882e16	-3.73170e14	6.38640e16	-1.89823e14	-21768070599475200
22	-1.49974e18	-4.24489e16	1.43795e18	-4.41050e16	2201197055016960000
26	-1.18673e22	-3.60961e21	1.18653e22	-3.61016e21	951354215990967140352000

Table 1: Periods $\omega^\pm(f_k)$ of $f_k = E_{k-12}\Delta$ and quasi-periods $\eta^\pm(\widehat{\Delta}_k)$ of the weak cusp form $\widehat{\Delta}_k$, and the integer $\det / (2\pi i)^{k-1}$ with $\det = \omega^+\eta^- - \omega^-\eta^+$. Full 50-digit values in [4].

2.1 Why a finite contour

Remark 2.8. For a weak cusp form $f \in S_k^!$ the classical Mellin integral $\int_0^\infty f(\tau)(\tau/i)^s \frac{d\tau}{\tau}$ does not converge — f grows at the cusp — and the deformed-Mellin construction of [1] supplies a finite value through a non-holomorphic regularization there. The functional of Definition 1.3 reaches the same value with no regularization: written as its two segments, each diverges as $\tau_0 \rightarrow i\infty$, with opposite signs, and the two cancel at every finite τ_0 , so $L^*(f, s)$ is finite throughout $\mathbb{H}$ although the Mellin contour itself has no limit. By Proposition 1.6 the value is independent of τ_0 and of the paths; BFK’s regularized value is its evaluation at $\tau_0 = i$ (Theorem 2.2), and its integer- s values are Brown’s quasi-periods (Remark 2.7). Two independent finite computations — Brown’s cohomological periods and the BFK L -values, obtained without reference to one another — meet at this one contour, which fixes it as the canonical object. Finiteness and τ_0 -independence are the analytic shadow of the parabolic representative [2, 6, 7].

3 Meromorphic modular forms

Definition 3.1 (Polylog projection). Let $f \in F_k$ have interior poles at the points τ_p of a set $\mathcal{P}_f$ of Γ -inequivalent representatives, and write $\text{Li}_s(z) = \sum_{n>0} z^n n^{-s}$. Since $\text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$ has a pole of order m at τ_p , the principal part of f there is matched by a unique polylog projection

$$\widehat{P}_{\tau_p} f(\tau) := \sum_{m>0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)}), \quad (19)$$

its coefficients $r_{f, \tau_p}^*(m)$ being the principal-part data of f at τ_p . The projection $\widehat{P} := \sum_{\text{im } \tau_p \geq 1} \widehat{P}_{\tau_p}$ removes the principal part of every pole on or above the line $\text{im } \tau = 1$.

Definition 3.2 (Regularized cusp coefficients). With $\widehat{P}$ as in Definition 3.1,

$$c_{Rf}(n) := [(I - \widehat{P})f]_{q^n}. \quad (20)$$

Definition 3.3 (The space $F(\infty)$). $F(\infty)$ is the space of meromorphic g on $\mathbb{H}$ with $g(\tau) = g(\tau+1)$, finitely many poles in any compact subset of the strip $\{\text{Im } \tau > 0, |\text{Re } \tau| \leq \frac{1}{2}\}$, and $g(\tau) = O(e^{C \text{Im } \tau})$ as $\text{Im } \tau \rightarrow \infty$ for some C . Each such g has a Fourier expansion $g = \sum_n c_g(n) q^n$.

Then $F_k \subset F(\infty)$, and $(I - \widehat{P})f \in F(\infty)$ — the polylog tails are T -periodic but not modular, so the projection leaves F_k .

Lemma 3.4 (*q-series-integral commutation*). *Let $g \in F(\infty)$ have no pole on or above the line $\text{Im } \tau = 1$ (e.g. $g = (I - \widehat{P})f$). Then $g = \sum_n c_g(n)q^n$ converges along the T -segment at $\tau_0 = i$ and integrates term by term:*

$$\int_{\gamma_i^T} g(\tau) \widetilde{k}_T(\tau, s) d\tau = \sum_n c_g(n) \int_{i-1}^i e^{2\pi i n \tau} \widetilde{k}_T(\tau, s) d\tau. \quad (21)$$

A pole at $\text{Im } \tau_p \geq 1$ defeats this — there the coefficients grow at least like $n^{m-1}e^{2\pi n \text{Im } \tau_p}$ [3, Lem. 2.6].

Proof. With no pole on or above $\text{Im } \tau = 1$, the finitely many poles of g in the strip have heights < 1 ; let $h < 1$ be the largest. The Fourier series $g = \sum_n c_g(n)q^n$ converges absolutely and uniformly on compact subsets of $\{\text{Im } \tau > h\}$, in particular on the T -segment $[i-1, i]$; with $\widetilde{k}_T(\tau, s)$ bounded there, the sum passes through the integral, which is the displayed identity. Conversely a pole at height $\text{Im } \tau_p \geq 1$ forces $|c_g(n)| \gtrsim n^{m-1}e^{2\pi n \text{Im } \tau_p}$ [3, Lem. 2.6], so $|c_g(n)q^n| \gtrsim n^{m-1}e^{2\pi n(\text{Im } \tau_p - 1)} \not\rightarrow 0$ on $[i-1, i]$ and the series diverges. $\square$

Definition 3.5 (*Polylog integral*). *For $N \geq 0$ and $z \in \mathbb{H}$, the polylog integral is the L -integral of a single tail $\text{Li}_{-N}(e^{2\pi i(\tau-z)})$ along the T -segment at $\tau_0 = i$ (where γ^S degenerates, since $-1/i = i$):*

$$I_N(s, z) := i^{-s} \int_{i-1}^i \text{Li}_{-N}(e^{2\pi i(\tau-z)}) \widetilde{k}_T(\tau, s) d\tau. \quad (22)$$

Lemma 3.6 (*Closed form of the polylog integral*). *Write $\mathcal{G}(s, m) := \frac{\Gamma(s, 2\pi m)}{(2\pi m)^s} + i^k \frac{\Gamma(k-s, 2\pi m)}{(2\pi m)^{k-s}}$, the building block of Theorem 2.2 continued to $m < 0$. For the projected poles $\text{Im } z > 1$, where $|e(\tau-z)| \geq 1$ on the contour, polylog inversion $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w)$ resums the integral into the convergent series*

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} \mathcal{G}(s, -n). \quad (23)$$

Proof. For $\text{Im } z > 1$, $|e(\tau-z)| \geq 1$ on $[i-1, i]$, so the series of $\text{Li}_{-N}(e(\tau-z))$ diverges; the inversion rewrites it through $\text{Li}_{-N}(e(z-\tau)) = \sum_{n \geq 1} n^N e^{2\pi i n(z-\tau)}$, convergent since $|e(z-\tau)| \leq 1$. Integrating term by term against $\widetilde{k}_T$, each $i^{-s} \int_{i-1}^i e^{-2\pi i n \tau} \widetilde{k}_T d\tau$ is the building block $\mathcal{G}(s, -n)$ of Theorem 2.2 at mode $-n$, giving the displayed sum; for $N = 0$ the leftover -1 in $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$ adds $-i^{-s} \int_{i-1}^i \widetilde{k}_T = \frac{1}{s} + \frac{i^k}{k-s}$. $\square$

Theorem 3.7 (*L -function of a meromorphic modular form*). *For $f \in F_k$ with no pole on the edge $\text{Im } \tau = 1$, at $\tau_0 = i$,*

$$L^*(f, s) = -c_{Rf}(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) \mathcal{G}(s, n) + \sum_{\text{Im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p), \quad (24)$$

$\mathcal{G}(s, n) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}$; the regularized cusp sum (Definition 3.2, Theorem 2.2) plus one polylog integral (Lemma 3.6) per interior pole above the edge. Poles on the edge enter by the principal value of Lemma 3.8.

Proof. At $\tau_0 = i$ the S -segment degenerates, so $L^*(f, s) = i^{-s} \int_{i-1}^i f \tilde{k}_T d\tau$. Split $f = (I - \hat{P})f + \hat{P}f$. The regularized part has no pole on or above $\text{Im } \tau = 1$ (Definition 3.1), so by Lemma 3.4 its Fourier series integrates term by term; mode by mode this is Theorem 2.2, giving $-c_{Rf}(0)(\frac{1}{s} + \frac{i^k}{k-s}) + \sum_{n \neq 0} c_{Rf}(n) \mathcal{G}(s, n)$. The projected part $\hat{P}f = \sum_{\tau_p} \sum_{m > 0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e(\tau - \tau_p))$ integrates to $\sum_{\text{Im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p)$ by Lemma 3.6. Their sum is the stated formula. $\square$

Lemma 3.8 (*L-function, a pole on the edge*). *Routing the T -path just below versus just above a pole τ_p of $f \in F_k$ at height $\text{Im } \tau_p = 1$ gives finite values differing by the residue,*

$$L_{\text{below}}^*(f, s) - L_{\text{above}}^*(f, s) = 2\pi i i^{-s} \text{Res}_{\tau=\tau_p} [f(\tau) \tilde{k}_T(\tau, s)]. \quad (25)$$

We define the edge value as the principal value $L^* = \frac{1}{2}(L_{\text{above}}^* + L_{\text{below}}^*) = L_{\text{below}}^* - \pi i i^{-s} \text{Res}_{\tau=\tau_p} [f \tilde{k}_T]$. At the elliptic point $\tau_p = i$ the pole lies on the S -segment instead; its principal value, reached as $\tau_0 \rightarrow i$, is finite and basepoint-independent.

Proof. The two routings of γ^T differ by a contour encircling τ_p once, so the residue theorem gives $L_{\text{below}}^* - L_{\text{above}}^* = 2\pi i i^{-s} \text{Res}_{\tau=\tau_p} [f \tilde{k}_T]$, and the principal value is their mean. At $\tau_p = i$ the pole sits on the degenerate S -segment; displacing to $\tau_0 = i(1 + \delta)$ keeps it on γ^S for all small $\delta > 0$, so no wall is crossed and L^* is finite and δ -independent (verified numerically [4]; a closed proof is the S -segment principal value of [3]). $\square$

Corollary 3.9 (Wall-crossing). *For $f \in F_k$, $L^*(f, s)$ is constant in τ_0 within each chamber of $\mathbb{H} \setminus f^{-1}(\infty)$; crossing a wall, where the contour sweeps past an interior pole τ_p , changes it by*

$$\Delta L^*(f, s) = 2\pi i i^{-s} \text{Res}_{\tau=\tau_p} [f(\tau) \tilde{k}_T(\tau, s)] \quad (26)$$

when the T -segment crosses τ_p (and by the same with τ^{s-1} in place of $\tilde{k}_T$ when the S -segment does). Thus L^* is fixed by the contour up to the residues it crosses — the meromorphic counterpart of Proposition 1.6, recovering its constancy on $M_k^!$.

Proof. Within a chamber the contour meets no pole, so the δ_T -inversion of Proposition 1.6 applies unchanged and $\partial_{\tau_0} L^* = 0$. Crossing a wall, the contour sweeps once past τ_p ; the residue theorem adds $2\pi i i^{-s} \text{Res}_{\tau=\tau_p} [f \tilde{k}_T]$ when the T -segment crosses it, or the same with τ^{s-1} when the S -segment does. $\square$

References

- [1] K. Bringmann, K.-H. Fricke, and Z. Kent, *Special L-values and periods of weakly holomorphic modular forms*, Ramanujan J. **24** (2011), 119–139.
- [2] F. Brown, *A class of non-holomorphic modular forms III*, arXiv:1710.07912 (2017).
- [3] D. A. McGady, *L-functions for meromorphic modular forms and sum rules in conformal field theory*, arXiv:1806.09874 (2018).
- [4] *Numerical verification scripts* (companion code), 2026.
- [5] W. Duke and P. Jenkins, *On the zeros and coefficients of certain weakly holomorphic modular forms*, Pure Appl. Math. Q. **4** (2008), no. 4, 1327–1340.

- [6] R. Bruggeman, Y. Choie, and N. Diamantis, *Holomorphic automorphic forms and cohomology*, Mem. Amer. Math. Soc. **253** (2018), no. 1212; arXiv:1404.6718.
- [7] K. Bringmann, P. Guerzhoy, Z. Kent and K. Ono, *Eichler-Shimura Theory for Mock Modular Forms*, Mathematische Annalen, March 2013 355(3).
- [8] W. Kohnen and D. Zagier, *Modular forms with rational periods*, in *Modular Forms* (R. A. Rankin, ed.), Ellis Horwood, 1984, pp. 197–249.
- [9] V. Paşol and A. A. Popa, *Modular forms and period polynomials*, Proc. London Math. Soc. (3) **107** (2013), 713–743; arXiv:1202.5802.
- [10] W. Wang and H. Zhang, *Meromorphic quasi-modular forms and their L -functions*, arXiv:2202.09542 (2022).
- [11] F. Brown and T. J. Fonseca, *Single-valued periods of meromorphic modular forms and a motivic interpretation of the Gross–Zagier conjecture*, arXiv:2508.04844 (2025).
- [12] K. Bringmann, N. Diamantis, and S. Ehlen, *Regularized inner products and errors of modularity*, Int. Math. Res. Not. IMRN (2016); arXiv:1603.03056.
- [13] N. Diamantis and J. Drewitt, *Period functions associated to real-analytic modular forms*, Res. Math. Sci. **7** (2020), no. 21.